THE MILLENNIUM PROBLEMS

BY

DOUGLAS QUIGG

RECONNAISSANCE
THE CORPUS, THE STATE OF THE FIELD, AND WHAT THIS TOOLKIT REACHES

ORIOR
2026

© 2026 Douglas Quigg
AGPL-3.0-or-later OR LicenseRef-Commercial OR LicenseRef-Educational

For Megan, my loving wife, the only person that really understands what I mean when I speak.

For Autumn and Opal, without you this thought would have never occurred.

To everyone else, I wish you well.

ABSTRACT

This is a reconnaissance and states no result. It records what the Millennium Prize problems ask, in the words of their official statements; what was published against one of them in September 2026; which of the rest an arbitrary-precision apparatus could be aimed at; and the reading that is not done. No problem is attacked here, and no priority is claimed over anything.

The corpus is seven documents. Five are Clay Mathematics Institute problem descriptions, covering P versus NP, the Hodge conjecture, the Riemann hypothesis, quantum Yang–Mills theory, and the Birch and Swinnerton-Dyer conjecture. Two are Riemann: his 1859 memoir in the Wilkins translation, and four photographed pages of his handwriting. Each was identified from its own pages. The photographs hold a working draft and not the published memoir, carry no extractable text, and circulate under a filename that names the memoir; a citation built from that filename would point at the wrong document.

Navier–Stokes is absent from the corpus because the Clay index of unsolved problems does not list it. On 8 September 2026 OpenAI published a proof of finite-time blowup for the forced three-dimensional incompressible equations, with a Lean 4 formalization. Fefferman's statement asks for a proof of any one of four statements, and two of the four name a smooth external force. Forcing is therefore inside the problem as posed.

The formalized statement is checked against the problem statement on six points: viscosity strictly positive with no vanishing limit, the decay conditions on the data and on the force, a force smooth on the closed half-line, a conclusion quantified as a negated existential, kinetic energy bounded for all time, and the periodic pressure condition the errata add. All six match. The statement is not the authors' own: it is adapted from the Formal Conjectures project's published formalization, and a separate module discharges it. A proof assistant checks a deduction and never a translation, and this arrangement answers that objection as far as an arrangement can.

The limits are stated where the claims are. The Lean development is not built and nothing here is machine-checked. Of the papers, the statements and the bridging argument are read and the proofs are not. A priority dispute runs in this field and no position is taken on it.

The problem the apparatus can reach is Birch and Swinnerton-Dyer. The refined conjecture is an exact numerical identity, and checking an instance comes down to whether a computed real number lands on an integer. The Riemann hypothesis is the trap: the apparatus matches its subject and not its obstruction, and counting more zeros bears on no proof. The null for the Birch and Swinnerton-Dyer measurement is drawn through the same pipeline as the effect, because a derived null sets the bar too low.

CONTENTS

1 WHAT THIS IS 1
1.1 The research paper 1
1.2 What is not claimed 1
1.3 Method 1

2 THE CORPUS 3
2.1 What is in it 3
2.2 The corpus is complete 3
2.3 The document that is not what it is named 4
2.4 An observation about production 4

3 NAVIER–STOKES, AND WHAT A FORMALIZATION CHECKS 5
3.1 What the problem asks 5
3.2 What was published against it 6
3.3 What a proof assistant does not check 6
3.4 The statement check 7
3.5 The statement is not the authors' own 7
3.6 The unforced Euler companion 8
3.7 What the unforced Euler result does not settle 8
3.8 What is not verified, stated here and not in a footnote 9

4 THE TERMS EQUATION (1) LEAVES OUT 11
4.1 The question 11
4.2 The balance laws, kept whole 12
4.3 The identity 12
4.4 What each condition asserts 13
4.5 A viscous flow makes its own temperature gradient 13
4.6 Dropping the gradient of viscosity 13
4.7 Condition (2) and thermal expansion 14
4.8 Where dropping the term is least safe 14
4.9 What keeping a magnitude does 15
4.10 Putting the fluid in a box 15
4.11 What this establishes and what it does not 16

5 WHAT THE AVERAGE KEEPS 19
5.1 The observation 19

5.2 The field and the molecules . . . 19
5.3 The average keeps a class and drops its members . . . 20
5.4 The equivalence is approximate at every finite count . . . 20
5.5 Where there is not enough volume or time . . . 21
5.6 What this establishes and what it does not . . . 21

6 WHAT THE FLUID HAS AND WHAT EQUATION (1) KEEPS . . . 23
6.1 The two sets . . . 23
6.2 Pressure at infinite speed . . . 23
6.3 Pressure with no level . . . 24
6.4 What this establishes and what it does not . . . 24

7 WHICH BOUNDARY WATER REACHES FIRST . . . 25
7.1 The coupled system . . . 25
7.2 Equation (1) cannot place the boundary . . . 26
7.3 The heat route . . . 26
7.4 The tension route . . . 27
7.5 Which comes first . . . 28
7.6 The heat route at the instant of contact . . . 28
7.7 What a blowup of (1) would have to show . . . 29
7.8 The heat of a collapsing core . . . 30
7.9 What this establishes and what it does not . . . 31

8 WHAT THIS TOOLKIT REACHES . . . 33
8.1 The obvious answer, and why it is the trap . . . 33
8.2 The recommendation . . . 33
8.3 The first measurement, and the positive control it requires . . . 34
8.4 The null, drawn and not derived . . . 34
8.5 The measured bound that constrains this . . . 35
8.6 What is not claimed . . . 35

9 SOURCES, AND HOW MUCH OF EACH WAS READ . . . 37
9.1 Read in full . . . 37
9.2 Read in part, with the part named . . . 37
9.3 Read at the first page only . . . 37
9.4 No text to read . . . 38
9.5 Not a document . . . 38
9.6 Cited for one fact each . . . 38
9.7 Named in these sources and not read here . . . 39
9.8 Availability . . . 39

BIBLIOGRAPHY 41

1

WHAT THIS IS

This chapter states what this research paper records and the boundary of what it claims.

1.1 THE RESEARCH PAPER

The aim is narrow: check one specific claim against its source documents and record what the check finds.

Four things are in it. First the corpus, and what each document is once opened, as against what its filename promises. Then the state of the Navier–Stokes problem as of September 2026. Then a reading of the published blowup result against the official problem statement and against the Lean formalization published with it. Last, an assessment of where an arbitrary-precision apparatus could be aimed among the remaining problems.

No problem is attacked. Nothing is measured, run, or begun. Leading a research paper with a method claims the method reaches something; no claim of that kind appears here or is supported here.

1.2 WHAT IS NOT CLAIMED

No priority is claimed. The crystallography research paper [30] says the same of its own construction.

The reading of the literature is not done. For any of the five open problems in the corpus, somebody has to work through the field behind it and set down what that field already knows. Until then, these pages assert no new method and credit no old one.

A priority dispute is running in the field the Navier–Stokes chapter covers. It is not investigated here, no position is taken, and no line in that chapter is evidence on either side. The chapter reports what the documents say and does not ask who reached the result first.

1.3 METHOD

Five rules govern the text, and each limits what the later chapters may claim.

Cite nothing unread. A document is cited here only once it is opened. A fact that arrives by report is marked as reported in the sentence that carries it.

A secondhand fact is no second witness. A fact that arrives secondhand is confirmed against the document before it is written down.

Gaps go in the sentence making the claim. Never in a footnote or an appendix. The Navier–Stokes chapter has a section on what is not verified, and marks the individual soft spots where they fall.

A claim that does not hold stays on the page, with what refutes it. The derived null in the toolkit chapter is kept beside the drawn null that replaces it.

Draw the bar. Never derive it. A threshold reasoned out of an assumed distribution omits every variance source not enumerated, and omitted variance only ever pushes one way. The toolkit chapter applies the rule to the recommended measurement.

2

THE CORPUS

This chapter states what each document in the corpus is, as established by opening it, and separates that from what its filename says. The corpus is seven files, 7,852,267 bytes.

2.1 WHAT IS IN IT

Each document is identified from its own first page and embedded metadata instead of its filename. Five are the official Clay Mathematics Institute problem descriptions [5, 7, 2, 18, 35]. Two are Riemann [36, 32].

File	Identified as	Pages	Bytes
`pvsnp.pdf`	Stephen Cook, *The P versus NP Problem*	12	176,204
`hodge.pdf`	Pierre Deligne, *The Hodge Conjecture*	5	146,894
`riemann.pdf`	E. Bombieri, *Problems of the Millennium: the Riemann Hypothesis*	11	159,267
`yangmills.pdf`	Jaffe and Witten, *Quantum Yang–Mills Theory*	14	195,976
`birchswin.pdf`	Andrew Wiles, *The Birch and Swinnerton-Dyer Conjecture*	5	142,908
`Wilkins-translation.pdf`	Riemann 1859, *On the Number of Prime Numbers less than a Given Quantity*, translated by David R. Wilkins	10	119,504
`riemann1859.pdf`	Not the 1859 paper. See below.	4	6,911,514

2.2 THE CORPUS IS COMPLETE

Six Millennium problems are commonly counted as open, and the corpus holds five official statements. The natural first reading is that Navier–Stokes is missing. That reading is wrong.

The Clay Mathematics Institute index of Millennium Prize problems [4], accessed 11 September 2026, lists five problems as unsolved, Birch and Swinnerton-Dyer, Hodge, P versus NP, Riemann, and Yang–Mills and the mass gap, and one as solved, Poincaré. Navier–Stokes appears in neither list. The corpus holds exactly the five the index lists as unsolved.

That is a fact about how a website categorizes its own contents on one date at one address, and it is recorded as that and not as a fact about the status of a problem. A site can be mid-edit, deliberately cautious, or behind. The Navier–Stokes problem's own page, accessed the same day, carries the status label *Active*. The two readings are not consistent with each other, both are recorded, and neither is treated here as evidence about the mathematics.

2.3 THE DOCUMENT THAT IS NOT WHAT IT IS NAMED

`riemann1859.pdf` is not Riemann's 1859 memoir. It is a photographic scan of Riemann's handwritten working manuscript [32].

The evidence is in the file. Its four pages carry zero extractable characters between them, measured page by page. The embedded document title is a scanner path, `Cod_Ms_Riemann_3_jpg`, under a directory named for the Riemann *Nachlass* and a subdirectory named *Rechenseiten*, calculation pages. The producer chain is a JPEG passed through PDF24 Creator and a Quartz PDF context in 2015, which is a scan-to-PDF path and not a typesetting one. Rendered and read as images, all four pages are German cursive in ink, heavily struck through with interlinear corrections, foxed and water-stained, with *Nachlass Riemann* and the year 1859 legible in the header of the first page. The zeta function, the factor $\Pi(s-1)$, the logarithmic expansion of the Euler product, the function $\xi(t)$ and the logarithmic integral $\mathrm{Li}(x)$ are identifiable in the hand.

The file is a primary source in the most literal available sense, and it is unusable for text search or for quotation without a reader of nineteenth-century German *Kurrentschrift*. The content of the published memoir is in the Wilkins translation [36] instead.

A citation built from this filename is wrong. It would name Riemann's published memoir and point at a manuscript draft. A bibliographic entry left with empty fields until somebody looks the source up does not catch the error, because the error is in the identification and not in the completeness. The distinction has to be written into the entry, and it is in this paper's.

2.4 AN OBSERVATION ABOUT PRODUCTION

Five of the six official statements, counting the Navier–Stokes statement [10] fetched separately for the next chapter, share a producer string of MiKTeX pdfTeX-1.21a and creation timestamps on 2026-08-04 within three minutes of one another; the Navier–Stokes file is stamped 16:46:57, nineteen seconds after the Birch and Swinnerton-Dyer file. Bombieri's Riemann statement comes from older tooling, a dvips and Acrobat Distiller chain dated 2000-08-30, and carries the embedded title `Riemann-new.dvi`.

The observation is about how the statements are produced and says nothing about their contents. It establishes that the corpus and the separately fetched statement are the same edition of the same set, which matters for a bibliographic entry and for no argument about mathematics.

3

NAVIER–STOKES, AND WHAT A FORMALIZATION CHECKS

This chapter states what the official problem asks, what was published against it on 8 September 2026, and the result of checking the published formalization's statement against the problem statement point by point. The documents are Fefferman's Clay problem description [10], 6 pages; the Navier–Stokes paper [28], 166 pages; the Euler paper [26], 57 pages; and the Lean 4 repository [27], of which four source files are read, one of them the adapted Formal Conjectures statement [11].

3.1 WHAT THE PROBLEM ASKS

Fefferman's official description sets up the incompressible equations on $\mathbb{R}^n$ for $n = 2$ or 3, with viscosity $\nu > 0$, and imposes decay and regularity conditions numbered (4) through (11). It then states, in the sentence that governs everything after it:

> To give reasonable leeway to solvers while retaining the heart of the problem, we ask for a proof of one of the following four statements.

The four are (A) existence and smoothness on $\mathbb{R}^3$, (B) the same on $\mathbb{R}^3/\mathbb{Z}^3$, (C) breakdown on $\mathbb{R}^3$, and (D) breakdown on $\mathbb{R}^3/\mathbb{Z}^3$. In (A) and (B) the force f is taken identically zero. In (C) and (D) it is not. Statement (C) reads, in his words, that for $\nu > 0$ and $n = 3$ there exist a smooth divergence-free u° on $\mathbb{R}^3$ *and a smooth* $f(x,t)$ on $\mathbb{R}^3 \times [0,\infty)$, satisfying (4) and (5), for which there exist no solutions (p,u) of (1), (2), (3), (6), (7) on $\mathbb{R}^3 \times [0,\infty)$. Statement (D) is the same with (8) and (9) on the data and (10) and (11) on the solution.

A smooth external force is therefore named inside two of the four statements, and the problem asks for a proof of any one of the four. The only occurrence of the word *prize* in the document is a remark that the Euler equations are not on the Clay Institute's list. There is no passage reserving the problem for (A) or (B).

The conditions that matter for the check below are (4), rapid decay of every derivative of the initial data; (5), rapid decay of every space-time derivative of the force on the closed half-line; (6), smoothness of p and u on $\mathbb{R}^n \times [0,\infty)$; and (7), $\int |u(x,t)|^2\,dx$ bounded by one constant for all $t \geq 0$, which Fefferman annotates *bounded energy*.

The document carries errata and one of them adds a condition. The first reads that the further condition $p(x + e_j, t) = p(x,t)$ should be made explicit in Eqn (8). The periodic problem therefore requires a periodic pressure, which the body of the statement does not say.

3.2 WHAT WAS PUBLISHED AGAINST IT

On 8 September 2026 OpenAI published *Finite time blowup for Navier–Stokes* [28], 166 pages, together with a companion Euler paper and a Lean 4 repository. Its Theorem 1.1 is a construction, not directly a non-existence statement:

> For every $\nu > 0$ there exist a force $f \in C_c^\infty(\mathbb{R}^3 \times (0,1); \mathbb{R}^3)$, a compact set $K \subset \mathbb{R}^3$, and smooth velocity and pressure fields u, p on $\mathbb{R}^3 \times [0,1)$ satisfying $\partial_t u + (u\cdot\nabla)u - \nu\Delta u + \nabla p = f$, $\nabla \cdot u = 0$, $u(\cdot,0) = 0$, such that $\operatorname{supp} u(\cdot,t) \cup \operatorname{supp} p(\cdot,t) \subseteq K$ for every $0 \le t < 1$, $\sup_t \|u(t)\|_{L^2(\mathbb{R}^3)} < \infty$, and $\limsup_{t\to 1} \|u(t)\|_{L^\infty(\mathbb{R}^3)} = \infty$.

Constructing one solution that blows up does not by itself establish (C). Statement (C) asks that *no* smooth bounded-energy solution exist on the whole half-line for that datum and force, and a blowing-up solution leaves open whether some other smooth global one exists. The paper closes that gap and says so in the sentence after the theorem, asserting that there is then no smooth solution on $\mathbb{R}^3 \times [0,\infty)$ with the same force and initial datum whose kinetic energy is uniformly bounded, and that this establishes alternative (C) as stated by Fefferman.

The argument for the gap is in the proof of Theorem 1.1 in section 10.4. Smoothness and fixed compact support put the field in $C([0,T]; H^3)$ for each $T < 1$; the embedding $H^3(\mathbb{R}^3) \hookrightarrow L^\infty(\mathbb{R}^3)$ excludes a classical H^3 continuation through time one; a difference-energy calculation with Grönwall gives uniqueness on every shorter interval, making $[0,1)$ the maximal classical existence interval. A supposed global smooth competitor of uniformly bounded energy is then forced by the comparison lemma to agree with the constructed field throughout $[0,1)$, which contradicts its own boundedness. Every fixed positive viscosity is reached by a spatial rescaling that leaves the singular time at one.

Statement (D) is Corollary 10.6, obtained by rescaling the compact supports into the interior of a fundamental cube and summing integer translates, whose supports stay disjoint so the nonlinear term survives. The corollary's proof states that the pressure is periodic as well, *as required in the erratum to the problem statement*. The paper accounts for the errata.

3.3 WHAT A PROOF ASSISTANT DOES NOT CHECK

The position taken here on machine-checked proof is that Lean verifies the deduction and never the claim. A proof assistant confirms that a conclusion follows from the assumptions as written. It cannot confirm that the statement formalized is the statement anybody meant. A machine-checked proof of a neighboring theorem compiles exactly as cleanly as a machine-checked proof of the right one, and nothing in the compile distinguishes them. The burden therefore moves entirely onto the formalized statement, and that statement is a translation. No proof assistant checks a translation.

That makes the formalized statement a finite document a person reads word by word. The rest of this chapter is that reading.

3.4 THE STATEMENT CHECK

Six points are checked against the Lean statement in `ComparatorChallenges/NavierStokes.lean` and against Fefferman's numbered conditions. The first five are specified before the file is opened. The sixth is added because the file itself raises it.

Point	What the Lean statement carries	Verdict
Viscosity strictly positive, and not a vanishing-viscosity limit	`(nu : R) (hnu : nu > 0)`, universally quantified over the theorem	matches
Condition (4) on the initial data, unrelaxed	`InitialVelocityConditionDecay`: divergence-free, C^∞, and for every derivative order m and every K a constant C with $\|\nabla^m u_0(x)\| \le C/(1+\|x\|)^K$	matches
Condition (5) on the force, unrelaxed, on the closed half-line	`ForceConditionDecay`: C^∞ on univ $\times [0,\infty)$ and $\|\nabla^m f(x,t)\| \le C/(1+\|x\|+t)^K$ for all $t \ge 0$	matches
Conclusion in the negated existential form Fefferman quantifies	`not (exists v p, NavierStokesExistenceAndSmoothnessRn nu u0 f v p)`	matches
Energy bounded uniformly in time, not at $t = 0$ alone	`globally_bounded_energy`: there exists E such that for all $t \ge 0$, $\int \|v(x,t)\|^2 < E$; plus square-integrability at each $t \ge 0$	matches
The periodic pressure condition from the errata	`isOnePeriodic_pressure` on the periodic structure, annotated in the source as following the Clay errata	matches

The equation itself is encoded with the time derivative taken as a derivative within $[0,\infty)$. Fefferman states equation (1) on the closed half-line, and the Lean encoding follows him there. Divergence is defined as the trace of the Jacobian and carries an explicit junk value of zero at points of non-differentiability, which the source declares instead of leaving implicit.

3.5 THE STATEMENT IS NOT THE AUTHORS' OWN

The file carrying the statement is a *comparator challenge*. Its header records that it is copied, at a named commit, from the Formal Conjectures project's Lean formalization of the Navier–Stokes problem statement [11], that the breakdown alternatives (C) and (D) are retained *including their intentional placeholder gaps*, and that neither the proof root nor the submission imports it. Both theorems in that file end in a placeholder and prove nothing. It is a yardstick.

The discharge is a separate module. It restates both theorems with textually identical statements and closes each by applying a bridge lemma from the project's own development, declaring no axioms and leaving no placeholder, and it ends by printing the axiom dependencies of both theorems. The repository's instructions direct a checker to verify the two challenges with the Comparator

tool under a sandbox, using an independent exporter and an independent kernel instead of Lean's own [27].

That architecture answers the objection this chapter opens with. The concern is that an author who writes both the statement and the proof can machine-check a translation of their own choosing. Here the statement is written by a third party, published beforehand, for the purpose of stating the problem and not of being proved; the authors adopt it unmodified and discharge it; and the axiom base is printed for a reader to check. The translation is still the thing to read, and this chapter reads it, but the party that wrote it has no stake in what is later proved against it.

3.6 THE UNFORCED EULER COMPANION

A second paper is published alongside, *Finite time blowup for the Euler equation* [26], 57 pages, and it matters here for one reason: it is unforced.

It studies the unforced incompressible Euler equations on $\mathbb{R}^3$, $\partial_t u + (u \cdot \nabla)u + \nabla p = 0$ with $\nabla \cdot u = 0$, taking initial data in the smooth compactly supported divergence-free class and writing $T(u_0)$ for the maximal lifespan of the smooth solution. Its Theorem 1.1 asserts that there exists such a u_0 with $0 < T(u_0) < \infty$, whose smooth solution satisfies $\limsup_{t \to T} \|u(t)\|_{L^\infty} = \infty$ and $\int_0^T \|\operatorname{curl} u(t)\|_{L^\infty}\, dt = \infty$.

The second divergence is not decoration. The Beale–Kato–Majda criterion, which the paper states and attributes, says a smooth three-dimensional Euler solution continues past T whenever that integral is finite. Finite-time breakdown therefore requires it to diverge, and the theorem delivers exactly the quantity the criterion demands. By the authors' own account the Euler result comes first, as the easier problem, and is used to prompt the Navier–Stokes effort.

This is not a Millennium result and is not recorded as one. Fefferman's statement says the problems are also open and very important for the Euler equations at $\nu = 0$, and that the Euler equation is not on the Clay Institute's list of prize problems. The distinction is his and is kept here.

3.7 WHAT THE UNFORCED EULER RESULT DOES NOT SETTLE

The question the Euler paper raises is what is left as the forcing goes to zero. The test behind it is whether a mechanism repairs a property of the object or a property of the setup; a mechanism that needs the force to be there teaches something about forced systems and nothing about the unforced problem.

The Euler paper looks like the answer and is not. **Removing the force and removing the viscosity are different removals.** Statements (A) and (B), the two alternatives that remain open on anybody's reading, are $\nu > 0$ with f identically zero. The Euler equations are $\nu = 0$. An unforced blowup at zero viscosity therefore does not transfer, and it does not transfer for the precise reason that the regularity problem exists: the viscous term is the smoothing whose sufficiency is the entire question. A construction that blows up with no viscosity to fight is not evidence about what happens when there is.

Put as the limit test, the two papers move in orthogonal directions. One removes the viscosity and keeps nothing to resist the amplification. The other keeps the viscosity and supplies a force. The open cases keep the viscosity and remove the force, and neither paper is that case.

That is an argument from the shapes of the statements and not from their mechanisms. The amplification mechanism is not read in either paper, and the full version of the limit test requires reading it instead of reasoning from the abstract. What is recorded here is that the question is still open and that the obvious shortcut to it does not work.

3.8 WHAT IS NOT VERIFIED, STATED HERE AND NOT IN A FOOTNOTE

Nothing here is machine-checked. The Lean development is not built. No proof is executed, no axiom output is observed, and the claim that the discharge module compiles without placeholders rests on reading its source text. The check in this chapter is a reading of four files out of 2,669.

The proofs are not read. The 166-page paper is read at its abstract, its introduction, its Theorem 1.1, the section 10.4 argument that bridges to the non-existence form, and Corollary 10.6. The mathematical content of sections 3 through 9 and the three appendices is not read. The bridge lemma the discharge module applies is not traced into the development. Of the Euler paper's 57 pages, only the abstract, the introduction and Theorem 1.1 are read; its amplification mechanism, the part the limit test needs, is not. Nothing in this chapter rests on the correctness of either proof, and nothing in it could, because neither is read.

The priority dispute is not investigated. A dispute over credit is running in this field. It is not examined here, no position is taken on it, and nothing in this chapter bears on it either way.

One fact rests on press reporting. That OpenAI does not intend to claim the Millennium Prize is reported in the press [33] and is not confirmed here against a statement of its own. It is recorded as reported.

And the status question is separate from the mathematics question. Whether the Clay Mathematics Institute accepts, awards, or will award anything is not verified here and is not asserted. What this chapter establishes is narrower and is the part that is checkable: the statement discharged in Lean is the statement Fefferman wrote, on all six points examined.

4

THE TERMS EQUATION (1) LEAVES OUT

This chapter shows that the equation Fefferman poses [10] is not a reduction of the balance laws of a real fluid but a different equation. It writes out every term the passage to it removes, proves that the removed terms are not zero, and shows what putting the fluid in a periodic box drops. It rests on equation (1) and condition (2) of Fefferman's statement, the continuum balance laws of mass, momentum and internal energy, the hypotheses of Theorem 1.1 of the Navier–Stokes paper [28] as the previous chapter quotes them, and the first five pages of Lienstromberg, Schiffer and Schubert [23].

4.1 THE QUESTION

Fefferman writes equation (1) in components,

$$\frac{\partial u_i}{\partial t} + \sum_{j=1}^{n} u_j \frac{\partial u_i}{\partial x_j} = \nu \Delta u_i - \frac{\partial p}{\partial x_i} + f_i(x,t),$$

with condition (2), $\nabla \cdot u = 0$. It carries one constant, ν. It carries no density and no temperature. The fluid is told its viscosity and is never asked for it.

A term may be dropped from an equation in two cases and only two: when it is zero, or when it is bounded by something small and the bound travels with the equation. This chapter writes the removed terms out in vector form, where each one is a single product that cannot be overlooked, and proves for each one that neither case holds for a fluid whose viscosity or density depends on its temperature. Water is such a fluid.

The fluid is taken to be a continuum at every scale, however small. So is Fefferman's: his u is smooth on $\mathbb{R}^3 \times [0,\infty)$. Nothing below departs from that. The fluid is also taken to fill all of $\mathbb{R}^3$ unless a container has been measured. Nothing is known about a container before its effect on the motion is read, and the last proved section shows what putting the fluid in a box costs.

This is the first observation of the thought experiment behind it. Every term is written in vector form, and nothing is kept fixed that the fluid has not been asked for. The observations that follow from it go in later chapters. The next one, *What the average keeps*, places the molecules in this field and bounds where an average of the two over a cell holds.

4.2 THE BALANCE LAWS, KEPT WHOLE

Write ρ for density, u for velocity, T for temperature, $\theta = \nabla \cdot u$ for the rate of volume change, and $D = \frac{1}{2}(\nabla u + \nabla u^{\mathsf{T}})$ for the rate of strain. The continuum balance of mass and of momentum are

$$\partial_t \rho + \nabla \cdot (\rho u) = 0, \qquad \rho\,(\partial_t u + (u \cdot \nabla)u) = \nabla \cdot \sigma + f,$$

where σ is the stress. The Newtonian stress law is the stress linear and isotropic in D,

$$\sigma = -p\,I + \lambda\,\theta\,I + 2\mu D,$$

with μ the shear viscosity and λ the second coefficient. Neither coefficient is a constant of nature. Each is a measured function of the state of the fluid, and the one this chapter uses is $\mu = \mu(T)$.

Where $\theta = 0$, the balance of internal energy is

$$\rho c\,(\partial_t T + u \cdot \nabla T) = \nabla \cdot (k \nabla T) + 2\mu\,|D|^2,$$

with c the specific heat, k the thermal conductivity, and $|D|^2 = \sum_{i,j} D_{ij}^2$. The last term is the heat that shear puts into the fluid, and it is never negative. These three balances are the standard ones and are not derived here.

4.3 THE IDENTITY

Proposition 1. For smooth u, p, μ and λ,

$$\nabla \cdot \sigma = -\nabla p + \mu\,\Delta u + (\lambda + \mu)\,\nabla\theta + \theta\,\nabla\lambda + 2D\,\nabla\mu.$$

Proof. In components, $\partial_j \sigma_{ij} = -\partial_i p + \partial_i(\lambda\theta) + \partial_j(2\mu D_{ij})$. The second term is $\lambda\,\partial_i\theta + \theta\,\partial_i\lambda$. The third is $2D_{ij}\,\partial_j\mu + 2\mu\,\partial_j D_{ij}$, and $2\partial_j D_{ij} = \Delta u_i + \partial_i\theta$. Collecting gives the statement. □

Set the density to one, θ to zero and $\nabla\mu$ to zero, and the momentum balance with the Newtonian stress law becomes equation (1) with $\nu = \mu$. The three terms that leave are

$$R = (\lambda + \mu)\,\nabla\theta + \theta\,\nabla\lambda + 2D\,\nabla\mu,$$

and equation (1) is exact at a point if and only if R is zero there.

In components the shear part alone of $\partial_j \sigma_{ij}$ expands to twelve terms for each i, and the three that carry $\partial_j\mu$ sit among nine that do not. In vector form the same three are one product, $2D\nabla\mu$. The notation does not make a term large. It makes a term impossible to drop without writing down the condition that drops it.

4.4 WHAT EACH CONDITION ASSERTS

Condition	Term it removes	What it asserts about the fluid
$\theta = 0$, condition (2)	$(\lambda + \mu)\nabla\theta + \theta\nabla\lambda$	no element of the fluid changes volume
$\nabla\mu = 0$	$2D\nabla\mu$	viscosity is the same at every point
$\rho \equiv 1$	the variation of ρ in the inertia	density is the same at every point and time
no energy balance	every coupling to T	temperature does not enter the motion
the stress law itself	any dependence of μ on $\lvert D\rvert$	stress is linear in the rate of strain

Condition (2) removes λ entirely. Whether the second coefficient is zero is therefore not answered by equation (1). It is removed from the question.

4.5 A VISCOUS FLOW MAKES ITS OWN TEMPERATURE GRADIENT

Proposition 2. Let u, T be smooth on $\mathbb{R}^3 \times I$ for an interval of time I, satisfy the energy balance above with $\theta = 0$, $\rho c > 0$ and $\mu > 0$, and let $D(x,t) \to 0$ as $|x| \to \infty$ for each t. If $\nabla T = 0$ everywhere on $\mathbb{R}^3 \times I$, then $u = 0$ on $\mathbb{R}^3 \times I$.

Proof. With $\nabla T = 0$ the temperature is a function of time alone, the conduction term vanishes, and the balance reads $\rho c\, T'(t) = 2\mu\, |D(x,t)|^2$ for every x. The left side does not depend on x. Letting $|x| \to \infty$ sends the right side to zero. Then $T'(t) = 0$, and $|D| = 0$ at every point. A smooth field with $D = 0$ is a rigid motion, $u = a(t) + W(t)x$ with $W^{\mathsf{T}} = -W$. Decay at infinity forces $a = 0$ and $W = 0$. □

Any motion of a viscous fluid that decays at infinity, motion with compact support among them, produces a temperature gradient within every interval of time it moves. The fluid writes its own temperature field. Equation (1) has nowhere to put it.

4.6 DROPPING THE GRADIENT OF VISCOSITY

Proposition 3. Let $\mu = \mu(T)$ and $\theta = 0$. The term removed by setting $\nabla\mu = 0$ is $2\mu'(T)\, D\, \nabla T$. It is zero at a point if and only if $\mu'(T) = 0$ there or ∇T lies in the null space of D. At a point where $\det D \neq 0$ and $\mu'(T) \neq 0$, it is zero if and only if $\nabla T = 0$.

Proof. The chain rule gives $\nabla\mu = \mu'(T)\nabla T$. A matrix with nonzero determinant has trivial null space. □

For water the factor $\mu'(T)$ is not small. IAPWS R12-08 [15], the formulation for the viscosity of ordinary water, evaluated by a program that reproduces its verification table to every printed digit, gives 1.0016 mPa s at 20 °C and atmospheric pressure and $\mu'/\mu = -0.0245$ per kelvin there: about two and a half percent per degree.

The term was dropped for a stated reason. Stokes [34], in his Article 5, holds μ constant as far as regards the temperature because, "in the case of small vibrations, the terms introduced by supposing it to vary with the temperature would involve the square of the velocity, which is

supposed to be neglected." The reason is a bound on the velocity, and it holds for small vibrations. Navier [24] reports from the experiments he cites that the action of the wall on the fluid diminishes as the temperature rises, and his equations hold his constants fixed.

A gap, stated here. Proposition 2 puts a nonzero ∇T somewhere in every interval of motion. Proposition 3 needs it at a point where $D\nabla T \neq 0$. Whether a flow can keep its own temperature gradient inside the null space of its own rate of strain at every point and time is not settled in this chapter.

4.7 CONDITION (2) AND THERMAL EXPANSION

Proposition 4. Let $\rho = \rho(T)$, let the mass balance and the energy balance hold, and let $\theta = 0$. At every point where $\rho'(T) \neq 0$,

$$\nabla \cdot (k \nabla T) = -2\mu \, |D|^2 \leq 0.$$

Proof. The mass balance with $\theta = 0$ is $\rho'(T)\,(\partial_t T + u \cdot \nabla T) = 0$. Where $\rho'(T) \neq 0$ the material rate of T is zero, and the energy balance leaves $0 = \nabla \cdot (k \nabla T) + 2\mu |D|^2$. □

Condition (2), taken together with a density that depends on temperature, therefore requires heat conduction to remove exactly the heat that shear puts in, at every point and every moment. A fluid that expands when heated does not do that. At a temperature where ρ' vanishes the proposition says nothing, and water has one: IAPWS R6-95 (2018) [17] puts its density maximum at 3.978 °C at atmospheric pressure. That is the only temperature at which condition (2) and thermal expansion agree without a further condition.

4.8 WHERE DROPPING THE TERM IS LEAST SAFE

Dropping $2D\nabla\mu$ claims that it is small, and D is the factor that an unbounded solution makes large.

Proposition 5. Let (u, p) be a smooth solution of (1) and (2) on $\mathbb{R}^3 \times [0, T^*)$ with $u(\cdot, t)$ supported in one compact set for every t, with f smooth and of compact support in $\mathbb{R}^3 \times (0, T^*)$, with $\sup_t \|u(t)\|_{L^2} < \infty$, and with $\limsup_{t \to T^*} \|u(t)\|_{L^\infty} = \infty$. These are the hypotheses of Theorem 1.1 of *Finite time blowup for Navier–Stokes* with $T^* = 1$. Write $Q(t) = \nu \int 2|D|^2 \, dx$, the rate at which shear puts heat into the fluid. Then

1. $Q(t) = \nu \, \|\nabla u(t)\|_{L^2}^2$;
2. Q has no finite bound on $[0, T^*)$;
3. $\int_0^{T^*} Q(t) \, dt < \infty$.

Proof. For the first, $\int 2|D|^2 = \int |\nabla u|^2 + \int \partial_j u_i \, \partial_i u_j$, and integration by parts with compact support gives $\int \partial_j u_i \, \partial_i u_j = -\int u_i \, \partial_i \theta = 0$.

For the second, the classical local theory gives, for data in H^1 with zero divergence and a smooth force, a smooth solution on an interval whose length has a positive lower bound depending only on $\|\nabla u\|_{L^2}$ at the start and on bounds of the force. If $\|\nabla u(t)\|_{L^2} \leq M$ on $[0, T^*)$, starting that theory at a time closer to T^* than the bound for M continues the solution smoothly past T^*, uniqueness makes the continuation agree with u, and $\|u\|_{L^\infty}$ stays bounded near T^*, which contradicts the

hypothesis. This step rests on the local theory, which is not reproduced here. Leray [22] states it in his Section 19: a solution regular before T that becomes irregular at T has $\max|u(\cdot,t)| > A\sqrt{\nu/(T-t)}$.

For the third, a smooth solution with compact support satisfies the energy equality

$$\tfrac{1}{2}\|u(t)\|^2 + \nu\int_0^t \|\nabla u\|^2 = \tfrac{1}{2}\|u(0)\|^2 + \int_0^t\!\!\int f\cdot u,$$

and the right side is at most $\tfrac{1}{2}\|u(0)\|^2 + \sup_s \|u(s)\|_{L^2}\int_0^{T^*}\|f(s)\|_{L^2}\,ds$. □

In the construction that settles statement (C), the heat that shear puts into the fluid arrives at a rate with no bound as the singular time approaches, while its total stays finite. Equation (1) has no channel through which that heat reaches the viscosity, and Proposition 3 says that channel is $2\mu'(T)\,D\nabla T$, with D the factor that has no bound. An argument that the dropped term is small needs a bound on each of its factors. Of the three, D is the factor equation (1) itself determines, and it has none.

What Proposition 5 does not show. It does not show that the temperature gradient becomes large in the collapsing region, that keeping the coupling stops the collapse, or that the coupled system is regular. The workbook carries a scaling estimate on the first of these, marked as theory.

4.9 WHAT KEEPING A MAGNITUDE DOES

The stress law can keep what equation (1) drops. A viscosity that depends on the magnitude of the rate of strain, $\mu(|D|) = \mu_0|D|^{p-2}$ with $p > 1$, puts a vector magnitude where equation (1) puts a constant, and $p = 2$ returns equation (1). Lienstromberg, Schiffer and Schubert [23] state that for $p > 11/5$ in three dimensions their approximations converge strongly to a solution that obeys the energy equality, that existence of solutions obeying an energy equality for such exponents traces back to Ladyzhenskaya [19, 20, 21], and that in the Newtonian case, $p = 2$, it is still not known whether a solution obeying an energy equality exists even for very regular initial data.

Which dependence the stress law keeps therefore changes which statements can be proved. This section shows that the choice decides the question. It does not show which stress law water obeys: the dependence of water's viscosity measured in this chapter is on temperature, not on $|D|$.

4.10 PUTTING THE FLUID IN A BOX

Statements (B) and (D) pose the problem on $\mathbb{R}^3/\mathbb{Z}^3$: the fluid in a periodic box of side one. Statement (D) is reached in the forced construction by placing the compact supports inside one cube and summing integer translates. Whether the box is a container the fluid is in, or one it is put in, decides what it keeps.

Write $\hat{u}(\xi) = \int_{\mathbb{R}^3} u(x)\,e^{-2\pi i\,\xi\cdot x}\,dx$, and for a rapidly decaying smooth u on $\mathbb{R}^3$ let $(Pu)(x) = \sum_{k\in\mathbb{Z}^3} u(x+k)$, the field put in the box.

Proposition 6. *Pu is smooth and periodic, and its Fourier coefficients are $\widehat{Pu}(m) = \hat{u}(m)$ for $m \in \mathbb{Z}^3$. The box therefore keeps $\hat{u}$ on the integer lattice and nothing between its points. There are nonzero smooth rapidly decaying fields u with $\nabla\cdot u = 0$ and $Pu = 0$. If u is supported inside one open unit cube, Pu equals u on that cube and nothing is lost.*

Proof. The coefficient is $\int_{[0,1)^3} \sum_k u(x+k)\, e^{-2\pi i\, m\cdot x}\, dx$, and since $e^{-2\pi i\, m\cdot k} = 1$ the sum over cells unfolds to $\int_{\mathbb{R}^3} u(x)\, e^{-2\pi i\, m\cdot x}\, dx = \hat{u}(m)$. A smooth periodic field is fixed by its coefficients, so $Pu = 0$ exactly when $\hat{u}$ vanishes on $\mathbb{Z}^3$. Take φ smooth, nonzero, supported in the two balls of radius $1/4$ about $\pm(1/2, 0, 0)$, with $\varphi(-\xi) = \overline{\varphi(\xi)}$ so that the field it makes is real. Let A be the field with $\hat{A} = \varphi\, e_3$, and set $u = \nabla \times A$. Then $\hat{u}(\xi) = 2\pi i\, \xi \times \hat{A}(\xi)$ is nonzero where φ is and vanishes on $\mathbb{Z}^3$, and $\nabla \cdot u = 0$. For a field supported in one open cube the translates do not overlap. □

Distinct flows on $\mathbb{R}^3$, as many as there are such φ, all enter the box as the same flow. The box loses nothing exactly when the fluid already fits inside one cell of it, and the forced construction uses that case. For a fluid larger than the box, the box keeps a lattice of samples of its transform and drops the rest.

Proposition 7. The rotations and reflections of $\mathbb{R}^3$ that fix the origin form an infinite group. Those that carry $\mathbb{Z}^3$ to itself, and so act on the box, number 48.

Proof. An orthogonal map carrying $\mathbb{Z}^3$ to itself sends each e_j to an integer vector of length one, which is $\pm e_k$. Such a map is a permutation of the three axes with a sign on each: $3! \cdot 2^3 = 48$. □

On $\mathbb{R}^3$ the expansion of a field in spherical harmonics about a point commutes with every rotation about that point, and the degree of a harmonic is then a property of the field. In the box only 48 of those rotations act. A wave pattern read in spherical harmonics on $\mathbb{R}^3$ is read against every direction; in the box it is read against three axes chosen before the fluid was looked at. This research paper does not show that spherical harmonics recover the container from the motion. That is a claim of the thought experiment, and the workbook holds it as theory.

4.11 WHAT THIS ESTABLISHES AND WHAT IT DOES NOT

Established. The passage from the balance laws to equation (1) removes the terms of Proposition 1 and the whole energy balance. For a fluid whose viscosity depends on temperature, the removed viscous term is zero only where the temperature gradient lies in the null space of the rate of strain. Any viscous motion that decays at infinity makes a temperature gradient. Condition (2), with a density that depends on temperature, forces conduction to cancel shear heating at every point. In the class of the forced construction, the rate of shear heating has no bound at the singular time. Putting a fluid larger than the box into it keeps its transform on a lattice and drops the rest, and the box keeps 48 of the rotations about a point. None of the dropped terms is zero, and none was dropped with a bound that travels with the equation.

Not established. Nothing here touches the correctness of the proof of Theorem 1.1. Its statements, the order in which its constants are chosen and its closing estimates were read; its technical bounds were followed and not checked line by line, and nothing read bears against it. A theorem about equation (1) is a theorem about equation (1), and the theorem stands as one. What this chapter bounds is what such a theorem is about: an equation from which the temperature has been removed, and not a fluid that has one. Questions (A) and (B) remain well posed questions about equation (1), and nothing here answers either.

The reasons for the dropped terms are stated where the equation was first written. Navier [24] takes for principle that the repulsion between two molecules is increased or diminished by a quantity proportional to the speed at which they approach or separate, and his fluid is incompressible from the start. Stokes [34] sets the bulk term to zero by assumption and writes that

agreement with experiment must not be regarded as confirming it; he holds μ constant in temperature for small vibrations, as above, and independent of pressure on Du Buat's experiments in pipes. Gibbs [12] gives the operators the propositions above use, and the split of a field into solenoidal and irrotational parts, unique on all of space.

The thought experiment that prompted this chapter reports a finding that a collapsing vortex in water would vaporize near 0.7 nanometers of width. No source for that report was located. The nearest source found, Duraiswami [9], puts the first failure of the continuum in water at cavitation near a millimeter of core radius, gives 0.3 nanometers as the molecular scale, and gives 0.7 micrometers only as the width at which the Knudsen number in air reaches 10^{-1}. Nothing here rests on the report.

5

WHAT THE AVERAGE KEEPS

This chapter shows that the field is continuous and the molecules sit in it, that equation (1) is a statement about averages of both over cells, that an average over enough volume and enough time gives an entropic equivalence that is approximate and never exact, and that a solution which breaks down takes away both the volume and the time. It rests on the second observation of the thought experiment, Proposition 5 of the previous chapter, and the density and viscosity of water from IAPWS R6-95 (2018) [17] and IAPWS R12-08 [15].

5.1 THE OBSERVATION

Take a quintillion molecules of a fluid. Each one feels the field and every neighbor, and each one moves on its own path. Equation (1) does not hold any of those paths. It holds a density, a velocity and, once the temperature is put back, a temperature: a few numbers at each point, standing for every molecule near that point.

5.2 THE FIELD AND THE MOLECULES

The field is defined at every point of $\mathbb{R}^3$. The molecules sit at a finite number of points in it. Nothing about the one rules out the other.

Proposition 8. Let Φ be a continuous field on $\mathbb{R}^3$ and let $x \neq y$ be any two points, two molecules among them.

1. If $\Phi(x) \neq \Phi(y)$, then on the segment from x to y the field takes every value between $\Phi(x)$ and $\Phi(y)$, and those values are uncountably many.

2. The map $g(t) = t/(1+t)$ carries $[0, \infty)$ one to one onto $[0, 1)$ and keeps order. Every finite value, however large, has a place in $[0, 1)$, and none reaches 1.

Proof. The function $s \mapsto \Phi(x + s(y - x))$ is continuous on $[0, 1]$, and the intermediate value theorem gives every value between its ends. An interval of real numbers with two distinct ends is uncountable. For the second, $g'(t) = 1/(1+t)^2 > 0$, $g(0) = 0$, $g(t) < 1$, $g(t) \to 1$, and $s \mapsto s/(1-s)$ is its inverse on $[0, 1)$. □

The differences of the field between two molecules are uncountably many, and every one of them is finite. Normalized to $[0, 1)$ they are all kept and none is cut off. The previous chapter is about this field, which is continuous at every scale. This chapter is about what equation (1) does with it: it averages the field, and every molecule in it, over a cell.

The average is in the definition of the velocity. Stokes [34], Article 1: "we may consider the mean velocity of the molecules ... apart from the velocity due to the irregular motion. It is this regular velocity which I shall understand by the velocity of a fluid at any point."

5.3 THE AVERAGE KEEPS A CLASS AND DROPS ITS MEMBERS

Take $N \geq 2$ molecules of mass m in one cell, with velocities $v_1, \dots, v_N$. The cell values the balance laws carry are the count N, the momentum $P = m\sum_i v_i$, and the kinetic energy $K = \frac{1}{2}m\sum_i |v_i|^2$. The count below is for the kinetic part alone. Energy stored in rotation, vibration and the forces between molecules adds to it.

Proposition 9. For fixed N, P and K with $K > |P|^2/(2Nm)$, the velocities that give those values form a sphere of dimension $3N-4$. With the positions free in the cell, the states of the N molecules that give the same cell values form a set of dimension $6N-4$.

Proof. Write $v_i = P/(Nm) + w_i$ with $\sum_i w_i = 0$. The cross term in K is $\frac{1}{N}P \cdot \sum_i w_i = 0$, and

$$K = \frac{|P|^2}{2Nm} + \frac{m}{2}\sum_i |w_i|^2.$$

The w lie in a subspace of dimension $3N-3$ and on a sphere in it of positive radius, which has dimension $3N-4$. The $3N$ coordinates of position add to it. □

For a quintillion molecules the cell keeps four numbers and drops a set of dimension $6\times10^{18}-4$. Every state in that set gives equation (1) the same values. The set is the equivalence class of the cell, and its size is what entropy measures: the Boltzmann entropy of the cell values is a constant times the logarithm of the size of the set. The average keeps the class. Which member of the class the molecules are in, which is every force on every molecule, is what it drops.

5.4 THE EQUIVALENCE IS APPROXIMATE AT EVERY FINITE COUNT

Proposition 10. Let $X_1, \dots, X_N$ be independent, each with mean a and squared spread $E[(X_i - a)^2] = s^2 > 0$, and let $\bar{X} = \frac{1}{N}\sum_i X_i$. Then $\bar{X}$ has mean a and squared spread s^2/N, which is positive for every finite N.

Proof. The squared spread of a sum of independent terms is the sum of the squared spreads of the terms, since every cross term $E[(X_i - a)(X_j - a)]$ with $i \neq j$ is zero, and scaling by $1/N$ scales it by $1/N^2$. □

The average equals the value it stands for only in the limit of infinitely many molecules. For a quintillion, its scatter is about 10^{-9} of the scatter of one molecule, and the continuum works in a glass of water for that reason. It is approximate there, and very good, and never exact.

What Proposition 10 assumes. It assumes independence. Molecules in a liquid are not independent: each one is constrained by its neighbors. Correlation between them makes the scatter larger, and the $1/\sqrt{N}$ rate is then the best case instead of a bound. The same shape applies to an average over time, with the time over which the motion stays correlated in the place of one molecule; that form is stated and not proved here.

5.5 WHERE THERE IS NOT ENOUGH VOLUME OR TIME

Proposition 11. In the class of Proposition 5, let $\ell(t) = \|u(t)\|_{L^2} / \|\nabla u(t)\|_{L^2}$, a length over which the velocity changes. Then ℓ has no positive lower bound on $[0, T^*)$, and neither does ℓ^2/ν, the time viscosity takes to act across ℓ.

Proof. $\ell(t) \le \sup_s \|u(s)\|_{L^2} / \|\nabla u(t)\|_{L^2}$. The numerator is finite by hypothesis, and Proposition 5 shows the denominator has no bound. □

Equation (1) carries no molecules, and Proposition 11 is a statement about equation (1) alone. Read as an average of a real fluid, it asks for cells that hold enough molecules and for a time over which to average that is long against the motion of one molecule and short against the motion of the flow. Proposition 11 takes both away. The length of the flow goes to zero, and with it the count of molecules in a cell of that length. The time of the flow goes to zero, and with it the window in which an average over time can be taken.

For water at 20 °C and atmospheric pressure, IAPWS R6-95 (2018) gives a density of 998.2072 kilograms per cubic meter, and with its specific gas constant and the Boltzmann constant that is 3.337×10^{28} molecules per cubic meter. A cube of side one micrometer holds 3.34×10^{10} of them, with a scatter under independence of 5.5×10^{-6}. A cube of side one nanometer holds 33.4, with a scatter of 17 percent. IAPWS R12-08 gives $\nu = 1.0034 \times 10^{-6}$ square meters per second, and ℓ^2/ν is one picosecond at that length. A solution of equation (1) that breaks down passes through those lengths and those times on its way, and there the average it is a statement about is no longer a good one.

5.6 WHAT THIS ESTABLISHES AND WHAT IT DOES NOT

Established. Equation (1), read as a law of a real fluid, is a law of cell averages. The average keeps an equivalence class of dimension $6N - 4$ and drops which member of it the molecules are in. Under independence, the average is approximate at every finite count and exact only in the limit. In the class of the forced construction, the length and the time of the flow have no positive lower bound. The cells and the windows the average needs do not stay large enough.

Not established. Nothing here says what the molecules do in a collapsing core, or that a description that keeps them would not break down. The scatter figures rest on independence, which a liquid does not have. Proposition 9 counts only the states of the molecules. The values the field takes between them, uncountably many by Proposition 8, are dropped by the same average and are not in that count.

6

WHAT THE FLUID HAS AND WHAT EQUATION (1) KEEPS

This chapter defines the set of every property of a fluid and the set of every force a fluid can feel, names the members equation (1) keeps, and proves two more ways in which what it keeps is not what the fluid has. It rests on equations (1) and (2) of Fefferman's statement [10], IAPWS R6-95 (2018) [17], the formulation for the thermodynamic properties of water, and Caupin and Herbert's review of cavitation in water [3].

6.1 THE TWO SETS

Definition. P_{all} is the set of all properties of a fluid, known and not yet known. F_{all} is the set of all forces a fluid can experience, known and not yet known. Both sets are open in the sense that no list of their members is claimed complete. A member enters a list when it is measured, and its absence from a list says only that it has not been measured yet.

Equation (1), with its density set to one, keeps

$$P_{(1)} = \{v\}, \qquad F_{(1)} = \{-\nabla p,\ \nu\Delta u,\ f\},$$

under the constraint $\nabla \cdot u = 0$. The term $(u \cdot \nabla)u$ is the rate of change of momentum carried with the flow and not a force.

Every member of $P_{\text{all}} \setminus P_{(1)}$ or $F_{\text{all}} \setminus F_{(1)}$ that is shown present by a measurement shows that equation (1) is a model on part of the two sets. The two sets being open, no finite list can show the reverse: that equation (1) keeps everything. The previous chapters exhibit members of both differences, and the workbook collects every one of them with the source that shows it present, as terms to put back.

6.2 PRESSURE AT INFINITE SPEED

Proposition 12. Let (u, p) be smooth and solve (1) and (2) on $\mathbb{R}^3$. At each instant,

$$\Delta p = \nabla \cdot f - \sum_{i,j} \partial_i \partial_j (u_i u_j).$$

Among solutions of this equation that decay at infinity, the one p is unique, and its value at any point depends on u and f at every point of $\mathbb{R}^3$ at the same instant.

Proof. Take the divergence of (1). By (2), $\nabla \cdot \partial_t u = \partial_t \theta = 0$ and $\nabla \cdot (\nu \Delta u) = \nu \Delta \theta = 0$. The remaining term is $\partial_i(u_j \partial_j u_i) = \partial_i \partial_j (u_i u_j) - \partial_i (u_i\, \partial_j u_j)$, and the second part is zero by (2). A harmonic function

on $\mathbb{R}^3$ that decays at infinity is zero, which gives uniqueness, and the decaying solution is the convolution of the source with $-1/(4\pi|x|)$, which is nonzero at every point. □

A change in the motion anywhere changes the pressure everywhere at the same instant. In water at 20 °C and atmospheric pressure, pressure travels at the speed of sound, 1482.35 meters per second, evaluated from IAPWS R6-95 (2018) by a program that reproduces its verification table to every printed digit. The speed of sound is a member of $P_{\text{all}} \setminus P_{(1)}$, and condition (2) sets it to infinity. The fundamental solution of the linear problem shows it directly: a force switched on at one instant gives a pressure proportional to $\delta(t)$ at every point of space [21].

6.3 PRESSURE WITH NO LEVEL

Proposition 13. If (u, p) solves (1) and (2), then $(u,\ p + c)$ solves them too, for every function c of time alone.

Proof. Pressure enters (1) only through ∇p, and $\nabla c = 0$. □

Equation (1) fixes no level of pressure, and conditions (6) and (7) of Fefferman's statement fix none either: neither bounds p at infinity. A real liquid has a level at which it breaks. Liquid water can be kept under tension, at negative pressure, until vapor bubbles form, and Caupin and Herbert report that classical nucleation theory puts that point at −168 MPa at 20 °C for the volume and time they state, that water in mineral inclusions is estimated to reach about −140 MPa, and that most other methods reach between a few and a few tens of megapascals below zero. Whether the liquid breaks depends on the level of the pressure. Equation (1) cannot pose the question, because it holds no level. Ladyzhenskaya [21] replaces the requirement that the pressure be non-negative by a bound on its integrals over surfaces, and writes that those integrals "only have physical meaning for areas Σ whose sizes are not less than a certain positive number (stipulated by the limits of accuracy of measurement and by the discreteness of the liquid medium)."

6.4 WHAT THIS ESTABLISHES AND WHAT IT DOES NOT

Established. Equation (1) keeps one property and three forces of the two open sets. It carries pressure at infinite speed, where water carries it at the speed of sound, and it holds no level of pressure, where water breaks at a level.

Not established. The sets are open, and nothing here bounds what they hold beyond the members measured. That keeping a member changes whether solutions break down is not shown for any member in this chapter.

7

WHICH BOUNDARY WATER REACHES FIRST

This chapter writes the system that equation (1) is cut from, with the viscosity a function of temperature and pressure, the energy balance with the heat of shear, and the boundary between liquid and vapor as a free surface. It proves that equation (1) cannot place that boundary, and it compares the two ways a moving body of water can reach it: by heating its liquid to the vapor point, or by lowering its pressure to the level at which the liquid breaks. It rests on Proposition 1 and Proposition 13, IAPWS R6-95 (2018) [17], R12-08 [15] and R15-11 [16], Caupin and Herbert's review [3], Gruntfest and Becker [14], and Duraiswami [9].

7.1 THE COUPLED SYSTEM

The fluid occupies a liquid region $\Omega_L(t)$ and a vapor region $\Omega_V(t)$, separated by a surface $\Gamma(t)$. In each region the balances of the chapter on the terms equation (1) leaves out hold, with the stress $\sigma = -p\,I + \lambda\,\theta\,I + 2\mu D$, and with every coefficient a measured function of the state:

$$\rho = \rho(T,p), \qquad \mu = \mu(T,p), \qquad \lambda = \lambda(T,p), \qquad k = k(T,p), \qquad c_p = c_p(T,p).$$

For water these are the IAPWS formulations. Without the condition $\theta = 0$, the balance of internal energy written for the temperature is

$$\rho\,c_p\,(\partial_t T + u\cdot\nabla T) = \beta\,T\,(\partial_t p + u\cdot\nabla p) + \nabla\cdot(k\nabla T) + \lambda\,\theta^2 + 2\mu\,|D|^2,$$

with $\beta = -\rho^{-1}\partial\rho/\partial T$ the thermal expansion coefficient. The last two terms are the heat of the motion, and neither is negative. These balances are the standard ones and are not derived here.

On Γ, let n be the unit normal pointing from liquid into vapor, w the speed of Γ along n, γ the surface tension and L the latent heat. The surface carries the balances

$$\dot m = \rho_L\,(u_L\cdot n - w) = \rho_V\,(u_V\cdot n - w),$$

$$\dot m\,(u_V - u_L) - (\sigma_V - \sigma_L)\,n = \nabla_\Gamma\gamma - \gamma\,(\nabla_\Gamma\cdot n)\,n,$$

$$\dot m\,L = -k_L\nabla T_L\cdot n + k_V\nabla T_V\cdot n,$$

with the temperature continuous across Γ and equal there to the saturation temperature at the vapor pressure. For a vapor bubble of radius R at rest the second balance gives $p_V - p_L = 2\gamma/R$, the law of Laplace.

The surface is empty until the liquid breaks somewhere. Write $p_{\mathrm{sat}}(T)$ for the vapor pressure and $\Delta_n \geq 0$ for the tension the liquid holds below it before a vapor bubble forms. A point of liquid stays liquid while

$$p > p_{\mathrm{sat}}(T) - \Delta_n.$$

The number Δ_n is a property of the liquid, of what is dissolved in it and of what it touches, and of the volume and time over which it is sustained. Caupin and Herbert report that classical nucleation theory gives 168 MPa for water at 20 °C, and that most methods reach a few to a few tens of megapascals.

Set Γ empty, the density and viscosity constant, and the energy balance aside, and the system is equation (1), by Proposition 1. Each of those steps drops a member of P_{all}, and the first drops the phase.

7.2 EQUATION (1) CANNOT PLACE THE BOUNDARY

Proposition 14. Let (u, p) solve (1) and (2) on $\mathbb{R}^3 \times I$, with $p(\cdot, t)$ bounded for each $t \in I$, and let p_* be any number. There are functions c_1 and c_2 of time alone such that $(u, p + c_1)$ and $(u, p + c_2)$ solve (1) and (2), the set $\{x : p(x,t) + c_1(t) \leq p_*\}$ is empty for every t, and the set $\{x : p(x,t) + c_2(t) \leq p_*\}$ is all of $\mathbb{R}^3$ for every t.

Proof. Take $c_1(t) = p_* + 1 - \inf_x p(x,t)$ and $c_2(t) = p_* - \sup_x p(x,t)$. Both pairs solve (1) and (2) by Proposition 13. □

Whether a point of the fluid lies past the level at which the liquid breaks is therefore not a property of a solution of (1). The same motion is liquid everywhere or past the breaking level everywhere, according to a constant equation (1) leaves free. The coupled system fixes that constant through $\rho(T, p)$ and the pressure the fluid is kept at.

7.3 THE HEAT ROUTE

Proposition 15. Let μ and k be positive continuous functions of T, and $F(T) = \int_{T_0}^{T} k(s)/\mu(s)\, ds$. Let the fluid fill the layer $|y| < \ell/2$ at a uniform pressure, with velocity $(U(y), 0, 0)$, walls kept at T_0 and moving at $\pm V/2$ with $V > 0$, and steady balances of momentum and energy,

$$(\mu(T)\, U')' = 0, \qquad (k(T)\, T')' + \mu(T)\, U'^2 = 0,$$

with $\mu U'$ and kT' continuously differentiable. If $V^2/8$ lies in the range of F, there is exactly one such solution. Its temperature is largest at $y = 0$, where it is the T_c with

$$V^2 = 8\, F(T_c) = 8 \int_{T_0}^{T_c} \frac{k(T)}{\mu(T)}\, dT,$$

which does not depend on ℓ.

Proof. The stress $\tau = \mu U'$ is constant, and positive since $V > 0$. The energy balance reads $(kT')' = -\tau^2/\mu(T) < 0$. Then kT' is strictly decreasing, and T has one largest value T_c, at the point y_c where $T' = 0$. Multiplying by kT' gives $\frac{1}{2}\big((kT')^2\big)' = -\tau^2\, F(T)'$, so that

$$(kT')^2 = 2\tau^2\big(F(T_c) - F(T)\big).$$

On each side of y_c, T is monotone and $dy = k\,dT/\big(\tau\sqrt{2(F(T_c)-F(T))}\big)$. The distance from y_c to each wall is the same integral from T_0 to T_c; hence $y_c = 0$. The velocity jump is

$$V = \int U'\,dy = 2\int_{T_0}^{T_c} \frac{\tau}{\mu}\,\frac{k\,dT}{\tau\sqrt{2(F(T_c)-F(T))}} = 2\int_0^{F(T_c)} \frac{dF}{\sqrt{2(F(T_c)-F)}} = 2\sqrt{2F(T_c)}.$$

Since F is continuous and strictly increasing, V fixes T_c. The width then fixes τ through $\ell = 2\int_{T_0}^{T_c} k\,dT/\big(\tau\sqrt{2(F(T_c)-F(T))}\big)$, an integral that converges because $F(T_c)-F(T)$ vanishes to first order at T_c. The relation for dy then gives $T(y)$, and $U' = \tau/\mu(T)$ gives U. □

The heat route needs a difference of velocity instead of a gradient of it: a thinner layer at the same V shears harder and conducts its heat out over a shorter distance, and its middle is no hotter. For constant μ and k the relation is $V^2 = 8k\,(T_c - T_0)/\mu$. For $\mu = \mu_0\,e^{-a(T-T_0)}$ and constant k it is $V^2 = (8k/(a\mu_0))\,(e^{a(T_c-T_0)} - 1)$, the relation of Gruntfest and Becker. Under that law the stress at a fixed width is proportional to $e^{-\varphi/2}\operatorname{arccosh}(e^{\varphi/2})$ with $\varphi = a(T_c - T_0)$. Its largest value is at $\varphi = 1.187$, the limit Gruntfest and Becker give for steady flow at a fixed stress. At a fixed velocity the solution continues past it, with the stress falling as V rises.

For water at atmospheric pressure, with $T_0 = 20\,^\circ\mathrm{C}$ and $\mu(T)$ and $k(T)$ from IAPWS R12-08 and R15-11, evaluated by programs that agree with their verification tables to six significant figures or more, k/μ rises from 597 to 2402 $\mathrm{m^2\,s^{-2}\,K^{-1}}$ between 20 and 99.9 °C. The middle of the layer stands at 20.012 °C at $V = 7.63$ m/s, at 20.060 °C at 16.9 m/s, at 36.7 °C at 315 m/s, and reaches 99.9 °C at 956 m/s.

7.4 THE TENSION ROUTE

Proposition 16. Let $u = v(r,t)\,e_\theta$ be a swirl about an axis with no radial or axial velocity, in a fluid of constant density ρ, with $p \to p_\infty$ as $r \to \infty$. Then at each t

$$p_\infty - p(0,t) = \rho\int_0^\infty \frac{v(r,t)^2}{r}\,dr.$$

For the Lamb–Oseen vortex, $v = (\Gamma_0/2\pi r)\big(1 - e^{-r^2/r_c^2}\big)$, an exact solution of (1) for any core radius $r_c(t)$ with $r_c^2 = r_0^2 + 4\nu t$,

$$p_\infty - p(0,t) = \rho\Big(\frac{\Gamma_0}{2\pi r_c}\Big)^2 \ln 2 = 1.702\,\rho\,v_{\max}^2,$$

with $v_{\max}$ the largest swirl.

Proof. With no radial velocity the radial component of the momentum balance is $\partial p/\partial r = \rho v^2/r$, and integrating from 0 to ∞ gives the first statement. For the Lamb–Oseen profile set $s = r/r_c$ and then $x = s^2$:

$$\int_0^\infty \frac{(1-e^{-s^2})^2}{s^3}\,ds = \frac{1}{2}\int_0^\infty \frac{(1-e^{-x})^2}{x^2}\,dx = \int_0^\infty \frac{e^{-x}-e^{-2x}}{x}\,dx = \ln 2,$$

by parts and then Frullani's integral. The swirl $(1-e^{-s^2})/s$ is largest at $s = 1.1209$, where it is 0.63817. Then $v_{\max} = 0.63817\,\Gamma_0/(2\pi r_c)$ and $\ln 2/0.63817^2 = 1.702$. □

For water at 20 °C, $\rho = 998.21$ kg/m^3 and $p_{\mathrm{sat}} = 2339$ Pa by IAPWS R6-95. From atmospheric pressure the axis of a Lamb–Oseen vortex reaches the vapor pressure at $v_{\max} = 7.63$ m/s, a tension of 1 MPa at 25.5 m/s, of 30 MPa at 133 m/s, and the 168 MPa of classical nucleation theory at 315 m/s.

7.5 WHICH COMES FIRST

Proposition 17. In water at 20 °C and atmospheric pressure, compare a layer of Proposition 15 and a vortex of Proposition 16 at the same speed, $V = v_{\max}$. Between 7.63 m/s, where the axis reaches the vapor pressure, and 315 m/s, where it reaches the tension of classical nucleation theory, the axis passes every breaking level Caupin and Herbert report, while the middle of the layer stays below 36.7 °C. At 7.63 m/s the middle of the layer is at most 0.0122 K above its walls, and its vapor pressure is raised by less than 1.8 Pa.

Proof. Since k/μ is increasing on the interval, $F(T_c) \geq (k/\mu)(T_0)\,(T_c - T_0)$, so that $T_c - T_0 \leq \mu(T_0)\,V^2/(8k(T_0))$, which at 7.63 m/s is 0.0122 K. The vapor pressure rises by 145 Pa per kelvin at 20 °C, and less than 1.8 Pa over that interval. The other figures are the computed values of the two previous sections. □

At a common speed the vortex lowers its axis pressure by $1.702\,\rho V^2$, and the layer raises the vapor pressure at its middle by about $(dp_{\mathrm{sat}}/dT)\,\mu V^2/(8k)$. Their ratio, $8 \cdot 1.702\,\rho k/(\mu\, dp_{\mathrm{sat}}/dT)$, is 5.6×10^4 for water at 20 °C. The tension route wins by that factor, and the boundary water reaches first is the cavity. This agrees with Duraiswami's computation for the forced construction, which puts its core at cavitation when the swirl reaches 10 to 17 m/s, with the heat of the remaining collapse near 0.01 K.

7.6 THE HEAT ROUTE AT THE INSTANT OF CONTACT

A layer between walls kept at T_0 loses its heat through the walls. Two sheets of one liquid brought together with a velocity jump have no walls at the boundary between them, and the heat of the first instant has nowhere to go but into the liquid on either side.

Proposition 18. Let one fluid of constant ρ, μ, c and k, at T_0, fill $y > 0$ moving at $V/2$ along x and $y < 0$ moving at $-V/2$, from $t = 0$. Write $\nu = \mu/\rho$, $\mathrm{Pr} = \mu c/k$ and $\eta = y/(2\sqrt{\nu t})$. The velocity $U = (V/2)\,\mathrm{erf}(\eta)$ and the temperature

$$T = T_0 + \frac{V^2}{4\pi c}\,G(\eta), \qquad G'(\eta) = -4\,\mathrm{Pr}\; e^{-\mathrm{Pr}\,\eta^2} \int_0^{\eta} e^{(\mathrm{Pr}-2)s^2}\,ds, \qquad G(\infty) = 0,$$

solve the balances of momentum and energy. For $\mathrm{Pr} > 2$ the temperature on the plane $y = 0$ is the same at every $t > 0$:

$$T(0,t) - T_0 = \frac{V^2}{2\pi c}\sqrt{\frac{\mathrm{Pr}}{\mathrm{Pr}-2}}\;\operatorname{artanh}\sqrt{1 - \frac{2}{\mathrm{Pr}}}.$$

Proof. The velocity is the solution of Stokes's first problem, $U_t = \nu U_{yy}$, with $U_y = (V/2)\,e^{-\eta^2}/\sqrt{\pi\nu t}$. The energy balance is $\rho c\,T_t = k\,T_{yy} + \mu U_y^2$, since nothing varies along x. With $T = T_0 + (V^2/4\pi c)\,G(\eta)$ each term carries the factor $1/t$, and dividing it out leaves

$$\frac{G''}{4\,\mathrm{Pr}} + \frac{\eta}{2}\,G' + e^{-2\eta^2} = 0.$$

The integrating factor $e^{\mathrm{Pr}\,\eta^2}$ and $G'(0) = 0$, from the symmetry of the two sides, give G' as stated. Then $G(0) = 4\,\mathrm{Pr}\int_0^\infty e^{-\mathrm{Pr}\,\eta^2}\int_0^\eta e^{(\mathrm{Pr}-2)s^2}\,ds\,d\eta$, and exchanging the order of integration,

$$G(0) = 2\sqrt{\pi\,\mathrm{Pr}}\int_0^\infty e^{(\mathrm{Pr}-2)s^2}\,\mathrm{erfc}\big(\sqrt{\mathrm{Pr}}\,s\big)\,ds = \frac{2\sqrt{\mathrm{Pr}}}{\sqrt{\mathrm{Pr}-2}}\ \mathrm{artanh}\sqrt{\frac{\mathrm{Pr}-2}{\mathrm{Pr}}},$$

by $\int_0^\infty e^{as^2}\mathrm{erfc}(bs)\,ds = \mathrm{artanh}(\sqrt{a}/b)/\sqrt{\pi a}$ for $0 < a < b^2$. The integral converges because $\mathrm{erfc}(bs)$ falls as $e^{-b^2s^2}$. □

The heat at the boundary is set by the difference in kinetic energy, $V^2/2$ per unit mass, divided by the heat capacity, times a number fixed by Pr alone. The viscosity sets how thick the heated layer is and not how hot it is, and neither the thickness of the sheets nor the time since contact enters. For water at 20 °C, $\mathrm{Pr} = 7.008$ and $c = 4184\,\mathrm{J\,kg^{-1}\,K^{-1}}$ by IAPWS R6-95, R12-08 and R15-11, and the boundary stands $5.578\times10^{-5}\,V^2$ kelvin above the sheets: 0.0032 K at 7.63 m/s, 0.56 K at 100 m/s, and 80 K, the boiling point at atmospheric pressure, at 1198 m/s. That speed is 0.81 of the speed of sound in water, where condition (2) and constant properties no longer describe it. At 7.63 m/s the tension route of Proposition 17 still comes first.

7.7 WHAT A BLOWUP OF (1) WOULD HAVE TO SHOW

A solution of (1) whose velocity has no bound describes a liquid near its singular time only if the liquid stays liquid there, its viscosity stays the constant (1) uses, and its density stays the constant (1) sets. Each of these is a condition on the temperature or on the level of the pressure.

Proposition 19. Let (u, p) be a smooth solution of (1) and (2) on $\mathbb{R}^3\times[0,T^*)$, bounded with its derivatives on $\mathbb{R}^3\times[0,t]$ for each $t < T^*$, and let T_b and p_* be numbers. Let T be the bounded solution of the energy balance carried by u,

$$\rho c\,(\partial_t T + u\cdot\nabla T) = k\,\Delta T + 2\mu\,|D|^2, \qquad T(\cdot,0) = T_b,$$

with ρc, k and $\mu = \rho\nu$ constant. Then $T \ge T_b$ at every point and time, and with c_2 from Proposition 14 the pressure $p + c_2$ is at most p_* at every point and time.

Proof. The difference $w = T - T_b$ starts at zero and solves $\rho c\,(\partial_t w + u\cdot\nabla w) - k\,\Delta w = 2\mu|D|^2 \ge 0$, a linear parabolic equation with bounded smooth coefficients on $\mathbb{R}^3\times[0,t]$. By the minimum principle for bounded solutions of such an equation, $w \ge 0$. The statement on the pressure is Proposition 14. □

Take T_b to be the boiling temperature at the pressure p_*. The solution (u, p) is then compatible with a fluid at or above its boiling point, at a pressure at or below p_*, at every point and every time: a liquid nowhere. Equation (1) is unchanged by the choice, because it holds no temperature and fixes no level of pressure. Whether the fluid stays liquid along a solution of (1) is therefore not a property of the solution. A theorem whose hypotheses and conclusions are statements about u and p cannot decide it.

To show that a blowup of (1) describes water, a proof would have to show three things near the singular point: that the temperature stays below boiling at the pressure there, that $2\mu'(T)\,D\nabla T$ stays small beside $\mu\Delta u$ where D has no bound (Propositions 3 and 5), and that the density holds (Proposition 4). Each needs the temperature and its gradient, and equation (1) has neither.

7.8 THE HEAT OF A COLLAPSING CORE

Proposition 19 holds for any solution. The construction Duraiswami computes [9] gathers its swirl into a core whose radius falls as $(T^* - t)^{1/2}$, and the heat of that swirl is the next question: how hot the core grows when its swirl is the construction's and its heat leaves by conduction alone.

Let a column of fluid of constant ρ, μ, c and k turn about the axis $r = 0$ with the swirl $v(r, t)$ and no other motion, and let the temperature depend on r and t alone. Then $u \cdot \nabla T = 0$, the one part of D is $D_{r\theta} = \frac{1}{2} r\, \partial_r (v/r)$, and the energy balance of Proposition 19 reads

$$\rho c\, \partial_t T = k\, \Delta T + \mu \left(r\, \partial_r (v/r) \right)^2 .$$

Write $\tau = T^* - t$, $X = r^2/(2\nu\tau)$, $\kappa = k/(\rho c)$ and $\mathrm{Pr} = \nu/\kappa$.

Proposition 20. Let $h \geq 0$ and let the swirl be $v = r\, \tau^{-1-h} F(X)$ on $0 \leq t < T^*$, with F continuously differentiable and $g(X) = \left(2XF'(X) \right)^2$ bounded and not zero everywhere. Let T be the bounded solution of the energy balance above with $T(\cdot, 0) = T_0$, and write $a = 1 + 2h$ and

$$\hat{g}(\lambda) = \int_0^\infty e^{-\lambda s}\, g(2s/\,\mathrm{Pr})\, ds .$$

Then on the axis, at every t in $(0, T^*)$,

$$T(0,t) - T_0 = \frac{\nu}{c}\, \tau^{-a} \int_{\tau/t}^\infty \frac{w^{a-1}}{(1+w)^a}\, \hat{g}(1+w)\, dw .$$

The number $\Theta_0 = \int_0^\infty w^{a-1} (1+w)^{-a}\, \hat{g}(1+w)\, dw$ is positive and finite, and

$$\frac{\nu}{c} \Big(\Theta_0\, \tau^{-a} - \frac{\hat{g}(1)}{a}\, t^{-a} \Big) \leq T(0,t) - T_0 \leq \frac{\nu}{c}\, \Theta_0\, \tau^{-a} .$$

The temperature on the axis has no bound as $t \to T^*$, and it grows as $(T^* - t)^{-1-2h}$.

Proof. Since $r\, \partial_r X = 2X$, $r\, \partial_r (v/r) = \tau^{-1-h}\, 2XF'(X)$, and the source is $Q = (\nu/c)\, \tau^{-2-2h} g(X)$. It is bounded on the plane for each $t < T^*$. The difference $T - T_0$ starts at zero, and by Duhamel's formula for the heat equation in the plane

$$T(0,t) - T_0 = \int_0^t \int_0^\infty \frac{e^{-r^2/(4\kappa(t-t'))}}{4\pi\kappa(t-t')}\, Q(r,t')\, 2\pi r\, dr\, dt' .$$

In the inner integral put $\tau' = T^* - t'$ and $s = \mathrm{Pr}\, r^2/(4\nu\tau')$. Then $X = 2s/\,\mathrm{Pr}$, $r\, dr = (2\nu\tau'/\,\mathrm{Pr})\, ds$ and $r^2/(4\kappa(t-t')) = s\, \tau'/(\tau' - \tau)$, and the inner integral is

$$\frac{\nu}{c}\, \frac{\tau'^{-1-2h}}{\tau' - \tau}\, \hat{g}\Big(\frac{\tau'}{\tau' - \tau} \Big) .$$

Now put $\tau' = \tau(1+w)/w$. Then $\tau'/(\tau' - \tau) = 1 + w$ and $dt' = \tau\, dw/w^2$, and as t' runs from 0 to t, w runs from τ/t to ∞. Collecting the powers of τ and w gives the formula. Every factor is positive, and the integral from τ/t is therefore at most Θ_0. On $0 < w < \tau/t$ the factor $(1+w)^{-a}$ is at most 1 and $\hat{g}(1+w)$ at most $\hat{g}(1)$, and the part of Θ_0 left out is at most $\hat{g}(1)\, (\tau/t)^a/a$. Last, $\hat{g}(\lambda) \leq \lambda^{-1} \sup g$. The integrand of Θ_0 is then at most $w^{a-1}(1+w)^{-a-1} \sup g$, which is integrable, and $\Theta_0 > 0$ because $g \geq 0$ is continuous and not zero everywhere. □

The power $-1-2h$ is twice the power of the peak swirl, $\tau^{-1/2-h}$. The temperature on the axis grows as the square of the fastest speed in the core, as the temperature at the instant of contact grows as V^2 in Proposition 18. The core's heat is made at the rate τ^{-2-2h}, and conduction moves it across the core's radius $(\nu\tau)^{1/2}$ in a time $\Pr\,\tau$, a fixed multiple of the time left. Conduction carries off a share of the heat that does not grow as the core shrinks, and the rest stays.

The outer swirl alone gives a bound in closed form.

Proposition 21. In Proposition 20 let $h = 0$ and let the swirl outside $X = X_b$ be the potential vortex, $v\,r = C$ for $X \geq X_b$, with C constant. Write $s_b = \Pr X_b/2$ and $E_1(s) = \int_s^\infty e^{-u}u^{-1}\,du$. Then at every t in $(0, T^*)$

$$T(0,t) - T_0 \geq \frac{C^2\,\Pr^2}{4\nu c}\Big(\frac{K(s_b)}{\tau} - \frac{e^{-s_b}/s_b - E_1(s_b)}{t}\Big), \qquad K(s) = \Big(1+\frac{1}{s}\Big)E_1(s) - \frac{e^{-s}}{s},$$

and $K(s) > 0$ for every $s > 0$.

Proof. For $X \geq X_b$, $F = \tau\,C/r^2 = C/(2\nu X)$. Then $2XF' = -C/(\nu X)$ and $g = C^2/(\nu^2X^2)$. The formula of Proposition 20 is linear in g with a positive kernel, and putting $g = 0$ inside X_b lowers it. With $a = 1$ the outer part is $\hat g_b(\lambda) = (C^2\,\Pr^2/4\nu^2)\int_{s_b}^\infty e^{-\lambda s}s^{-2}\,ds$, and since $\int_0^\infty e^{-(1+w)s}(1+w)^{-1}\,dw = E_1(s)$, exchanging the order of integration gives

$$\int_0^\infty \frac{\hat g_b(1+w)}{1+w}\,dw = \frac{C^2\,\Pr^2}{4\nu^2}\int_{s_b}^\infty \frac{E_1(s)}{s^2}\,ds.$$

Two integrations by parts with $E_1' = -e^{-s}/s$ give $\int_{s_b}^\infty E_1(s)\,s^{-2}\,ds = K(s_b)$, and one gives $\int_{s_b}^\infty e^{-s}s^{-2}\,ds = e^{-s_b}/s_b - E_1(s_b)$, which is $\hat g_b(1)$ up to the same factor. The bound of Proposition 20 with $\hat g_b$ in place of $\hat g$ is the statement. For the sign, $K(s) > 0$ is $E_1(s) > e^{-s}/(1+s)$. Write $E_1(s) = e^{-s}s^{-1}\int_0^\infty s\,e^{-su}(1+u)^{-1}\,du$: the integral is the mean of the convex function $1/(1+u)$ under the density $s\,e^{-su}$, whose mean is $1/s$, and by Jensen's inequality it is above $1/(1+1/s)$, strictly because the density is not concentrated at a point. □

The constant C/ν is the circulation of the outer swirl divided by $2\pi\nu$. The bound grows as its square and falls as $K(s_b)$, which is near e^{-s_b}/s_b^3 for large s_b: the two terms of K agree in their leading part, and their difference is what conduction leaves on the axis. Duraiswami's construction takes $0 < h < 1/100$ and joins its core to its outer swirl in $1 < X < 2$. Its outer swirl at $h = 0$ is the potential vortex, and Proposition 21 with $X_b = 2$ and $s_b = \Pr$ is that case.

7.9 WHAT THIS ESTABLISHES AND WHAT IT DOES NOT

Established. The system equation (1) is cut from has a phase boundary, and equation (1) cannot place it, because the boundary depends on a level of pressure that equation (1) leaves free. In steady shear between walls at a fixed temperature, the heat of shear sets the middle temperature by the velocity difference alone, for any viscosity and conductivity that depend on temperature. In water at 20 °C and atmospheric pressure, a turning column reaches the vapor pressure on its axis at a speed at which a shearing layer is warmed by a hundredth of a kelvin. Two sheets of one liquid brought together with a velocity jump V heat their boundary at once, by an amount set by V^2, the heat capacity and the Prandtl number, and the same at every later instant. Every solution of (1), carried with a temperature, is compatible with a fluid past its vapor point everywhere. Whether

a blowup of (1) describes a liquid cannot be decided in (1). A column turning with a swirl that collapses like the construction's has no bound on the temperature of its axis before its swirl has none, and that temperature grows as the square of the peak swirl.

Not established. The comparison is between two exact solutions at the same speed. It does not show that the forced construction, or any collapsing solution, follows either profile. The column of Propositions 20 and 21 has no flow along or toward its axis, its swirl is given and does not solve (1), and its coefficients are fixed at their values at T_0. The construction's inflow carries heat that the column leaves out, and nothing here says whether it carries that heat toward the axis or away from it. It treats the liquid as one substance. A boundary between two liquids that differ lowers Δ_n and the boiling point there, and the workbook holds what that predicts in a laboratory as theory. Nothing here shows that the coupled system is regular, or that its solutions break down.

8

WHAT THIS TOOLKIT REACHES

This chapter assesses which problem an arbitrary-precision apparatus could be pointed at, the reasoning for it, the trap beside it, and the measured bound that constrains it. The assessment is a recommendation and no work on it is begun.

8.1 THE OBVIOUS ANSWER, AND WHY IT IS THE TRAP

An apparatus whose edge is arbitrary precision reaches for Riemann first, and Riemann is the wrong choice for a reason that has nothing to do with how good the arithmetic is.

The apparatus matches the *subject* and not the *obstruction*. A zeta function and unbounded precision look like a fit. But zeros on the critical line are computed far past anything a desktop reaches. The first 10^{13} zeros are located on the line [13], zeros near the 10^{22}nd are computed [25], and the hypothesis is verified rigorously up to height $3 \cdot 10^{12}$ [29]. No finite count of zeros bears on the proof at all. Adding zeros reproduces existing work at a worse scale. The conjecture is not precision-bound, and an apparatus whose advantage is precision buys nothing against it.

That is the superficial-match failure in its exact form, and it is worth naming because it is the shape of mistake this apparatus is most exposed to: the subject looks like the instrument. Yang–Mills wants constructive quantum field theory and lattice gauge machinery. Hodge wants algebraic geometry with no numerical handle this toolkit has. P versus NP is not a numerical question.

8.2 THE RECOMMENDATION

Birch and Swinnerton-Dyer, because its obstruction matches the instrument, where the other Millennium problems match only in subject.

It is the only one of them found by computation: Birch and Swinnerton-Dyer fitted curves numerically on EDSAC [1]. It is natively experimental, and the refined conjecture is an exact numerical identity [35]. For an elliptic curve E of rank r, writing Sha for the order of the Tate–Shafarevich group, Ω for the real period, Reg for the regulator and c_p for the Tamagawa numbers,

$$\frac{L^{(r)}(E,1)}{r!} = \frac{\mathrm{Sha} \cdot \Omega \cdot \mathrm{Reg} \cdot \prod_p c_p}{|E_{\mathrm{tors}}|^2}.$$

Rearranged, the analytic order of Sha is a quotient of computable real numbers, and the conjecture asserts that this quotient is an integer, usually a perfect square. The verification is therefore: compute a real number to N digits and ask whether it lands on an integer. That is a precision-bound question, and it is the check the arithmetic already performs.

8.3 THE FIRST MEASUREMENT, AND THE POSITIVE CONTROL IT REQUIRES

The positive control comes first and is not optional. Curves are taken from published tables [6] where the rank and the order of Sha are already known, at rank zero and one, small conductor, with Sha trivial or known to be 4 or 9. The quotient is computed and must land on the known integer. An instrument that cannot reproduce a known answer says nothing when pointed at an unknown one.

The quotient is only as good as its weakest input. The real period and the regulator come from their own algorithms carrying their own error. The report is therefore the precision of the quotient and not the precision of the arithmetic. An arithmetic with no floor under it does not give its inputs one.

8.4 THE NULL, DRAWN AND NOT DERIVED

The derived null does not hold. It assumes that a real number carrying no integrality structure has its distance to the nearest integer distributed uniformly on $[0, \frac{1}{2}]$, that landing within 10^{-N} of an integer by chance therefore has probability about 2×10^{-N}, and that agreement to fifty digits is a bound of 10^{-50}.

What refutes it. It is derived and not drawn. A threshold reasoned out from an assumed distribution omits the variance sources not enumerated, and omitted variance only ever pushes one way.

One narrower objection to it is itself wrong. It holds that the integer factors in the quotient, the torsion order squared, the Tamagawa product and $r!$, might concentrate the quotient near integers and inflate the coincidence rate. They do not: scaling a structureless real by a known rational does not concentrate it near integers. The derived null fails for the better reason and not for that one.

The drawn null measures the null under the same conditions as the effect, through the same pipeline and with the same inputs and the same precision losses, in two ways. Forming the quotient with the rank taken one too high or one too low gives a condition under which the identity should not hold, and the distribution of distance-to-nearest-integer across many curves is then the null itself. Computing the quotient across many curves whose Sha is known gives the spread of the residual about the true integer. That spread is the instrument's own floor, and the period and the regulator dominate it well before the arithmetic does. The second is the positive control and the floor measurement in one pass.

The consequence changes what may be claimed. The bound is not 10^{-N} for N digits of arithmetic. It is however close the pipeline gets on a curve whose answer is already known. If the regulator runs out at 10^{-12}, then 10^{-12} is the claim however many digits the multiply carries.

8.5 THE MEASURED BOUND THAT CONSTRAINS THIS

The cost of depth in the integer relation search of the same arithmetic [31] is measured, and the measurement constrains the plan.

Cost goes as precision to the power 2.64. The prediction is 2.585, from a Karatsuba host multiply near width$^{1.585}$ against a count linear in the width, and the reduction step does not account for the difference.

The device overtakes the host at 1,024 limbs, 32,768 bits. The mean operand runs at 0.642 of the precision, and the mean multiply reaches that crossover near 51,000 binary places. One cell at 51,036 places costs 26.2 hours.

None of that binds the recommended measurement, which wants tens to hundreds of digits. It binds two other things: relation detection at tens of thousands of digits is out of reach, and so is any plan that relies on the device to rescue it.

8.6 WHAT IS NOT CLAIMED

This is verification and not proof, and that distinction governs everything claimed in this chapter.

A measurement of this kind produces evidence for particular instances of the conjecture, in the way the original EDSAC work did. A verified instance at unusual rank or unusual precision is a real contribution and it is not a solution.

The recommendation carries plain caveats. No expertise in elliptic-curve L-function computation stands behind it. Computing $L^{(r)}(E, 1)$ requires modular symbols [6] or Dokchitser's algorithm [8], established systems already implement those, and rebuilding them would be wasted effort. The realistic shape is established algorithms for the L-function with this arithmetic supplying precision underneath, and the first task is finding out whether that seam works at all.

9

SOURCES, AND HOW MUCH OF EACH WAS READ

This chapter names every source the research paper rests on and states how much of it is read, so that no claim here traces to a document nobody opened.

A surname in a sentence buys a reader nothing; an author, a title and an identifier they can follow buy them the source. Fields a document does not carry stay empty, with a note naming which. The extent of reading belongs beside each entry. Five sources are opened to the first page, two are read in part with the part named, one is read in full, and one cannot be read as text. Without that, a bibliography degrades into a list of things that exist.

9.1 READ IN FULL

Fefferman [10], *Existence and Smoothness of the Navier–Stokes Equation.* Six pages, errata included, read cover to cover; the Navier–Stokes chapter turns on it. No year appears anywhere on it, and the paper examined in that chapter cites it without one too; the entry carries none. Its bytes are identical to the copy claymath.org serves, accessed 11 September 2026.

9.2 READ IN PART, WITH THE PART NAMED

Finite time blowup for Navier–Stokes [28]. 166 pages. Abstract, introduction, Theorem 1.1, the bridging argument in section 10.4, Corollary 10.6. Sections 3 through 9 and all three appendices carry the mathematics and are not read.

Finite time blowup for the Euler equation [26]. 57 pages. Abstract, introduction, Theorem 1.1. The amplification mechanism is in the unread remainder, and the limit test in the Navier–Stokes chapter wants that mechanism.

Formal Conjectures [11], the Lean formalization of the Navier–Stokes problem statement. Read as source text, in the adapted copy the Lean repository [27] carries, at the commit that file names. Four files out of 2,669, none of them built. Everything this research paper finds about the formalized statement rests on that one file.

9.3 READ AT THE FIRST PAGE ONLY

Five official Clay problem descriptions, for the five problems the index lists as unsolved. Each is opened, identified from its own first page instead of its filename, and closed. All five are byte-identical to the copies claymath.org serves, accessed 11 September 2026.

Cook [5], *The P versus NP Problem*, 12 pages.

Deligne [7], *The Hodge Conjecture*, 5 pages.

Bombieri [2], *Problems of the Millennium: the Riemann Hypothesis*, 11 pages. Older tooling than the other four.

Jaffe and Witten [18], *Quantum Yang–Mills Theory*, 14 pages.

Wiles [35], *The Birch and Swinnerton-Dyer Conjecture*, 5 pages.

One page of Wiles is the largest shortfall here, because Birch and Swinnerton-Dyer is the problem the toolkit chapter recommends. That recommendation stands on an exact numerical identity, and the identity is not taken from this document. Whoever acts on the recommendation reads this first.

Wilkins [36], Riemann's memoir translated by David R. Wilkins, *On the Number of Prime Numbers less than a Given Quantity*, 10 pages. The year is the translation's, printed there as a preliminary version of December 1998; the memoir's date, November 1859, is printed beside it. English, legible, and the practical way into the 1859 content.

9.4 NO TEXT TO READ

Riemann 1859 [32], handwritten working manuscript, *Nachlass* Cod. Ms. Riemann 3, *Rechenseiten*. Four pages with zero extractable characters, counted page by page, rendered and read as images, the only way they can be read. The year on the entry is the manuscript's; the scan dates from 2015. Its filename names the published memoir and it holds a draft, and the corpus chapter states what a citation built from that filename would do.

9.5 NOT A DOCUMENT

Clay Mathematics Institute [4]. Two facts in this research paper describe what that website says on 11 September 2026. The Millennium Problems index lists five unsolved, one solved, and Navier–Stokes in neither list. The Navier–Stokes page itself carries the label *Active*. The two do not agree. Both are recorded as statements about how a website sorts its own contents on one day at one address, and neither is used as a statement about a problem's status.

9.6 CITED FOR ONE FACT EACH

The computations of Riemann zeros [13, 25, 29], the Birch and Swinnerton-Dyer computation [1], the curve tables [6], Dokchitser's algorithm [8] and the press report on the prize [33] are each cited for the single fact the toolkit or Navier–Stokes chapter takes from them, as confirmed against the publisher or an index, and not for their content beyond it.

9.7 NAMED IN THESE SOURCES AND NOT READ HERE

Reference lists carry names, and this research paper repeats a handful at second hand. Leray on global weak solutions. Caffarelli, Kohn and Nirenberg on partial regularity. The Beale–Kato–Majda continuation criterion. Buckmaster and Vicol on non-uniqueness, then Albritton, Brué and Colombo on the same. Elgindi on axisymmetric blowup.

None is read, and none is cited here as established. Each has the status of a thing a read source mentions, which is all it has earned. An entry for any of them would assert a reading that is not done.

9.8 AVAILABILITY

The Clay statements, the two papers and the Lean repository are public at the addresses in the bibliography. The arbitrary-precision arithmetic behind the toolkit chapter is described in the instruments research paper [31].

BIBLIOGRAPHY

[1] B. J. Birch, H. P. F. Swinnerton-Dyer. *Notes on elliptic curves. II.* Journal für die reine und angewandte Mathematik 218, 79-108. 1965.

[2] E. Bombieri. *Problems of the Millennium: the Riemann Hypothesis.* Clay Mathematics Institute, Millennium Prize Problem statement, https://www.claymath.org/wp-content/uploads/2022/05/riemann.pdf.

[3] F. Caupin, E. Herbert. *Cavitation in water: a review.* Comptes Rendus Physique 7(9-10), 1000-1017, doi:10.1016/j.crhy.2006.10.015. 2006.

[4] Clay Mathematics Institute. https://www.claymath.org/millennium-problems/.

[5] Stephen Cook. *The P versus NP Problem.* Clay Mathematics Institute, Millennium Prize Problem statement, https://www.claymath.org/wp-content/uploads/2022/06/pvsnp.pdf.

[6] J. E. Cremona. *Algorithms for Modular Elliptic Curves.* Second edition, Cambridge University Press. 1997.

[7] Pierre Deligne. *The Hodge Conjecture.* Clay Mathematics Institute, Millennium Prize Problem statement, https://www.claymath.org/wp-content/uploads/2022/06/hodge.pdf.

[8] T. Dokchitser. *Computing special values of motivic L-functions.* Experimental Mathematics 13(2), 137-150. 2004.

[9] R. Duraiswami. *Self-similar swirl between contracting porous walls: the GD1998 exact Navier–Stokes solution revisited in the similarity variables of the OpenAI 2026 forced blow-up construction.* arXiv:2609.17642v1. 2026.

[10] Charles L. Fefferman. *Existence and Smoothness of the Navier-Stokes Equation.* Clay Mathematics Institute, Millennium Prize Problem statement, https://www.claymath.org/wp-content/uploads/2022/06/navierstokes.pdf.

[11] Google DeepMind Formal Conjectures authors. *Lean formalization of the Navier-Stokes existence and smoothness problem statement.* https://github.com/google-deepmind/formal-conjectures/blob/main/FormalConjectures/Millenium/NavierStokes.lean.

[12] J. W. Gibbs. *Elements of Vector Analysis, arranged for the use of students in physics.* New Haven, printed 1881–1884 and never published. 1884.

[13] X. Gourdon. *The 10^{13} first zeros of the Riemann Zeta function, and zeros computation at very large height.* 2004.

[14] I. J. Gruntfest, S. J. Becker. *Mechanics of Deformation and Fracture.* Final report on NASA contract NASw-708, General Electric Re-entry Systems Department, NTRS 19650001075. 1964.

[15] International Association for the Properties of Water and Steam. *Release on the IAPWS Formulation 2008 for the Viscosity of Ordinary Water Substance*. IAPWS R12-08, Berlin. 2008.

[16] International Association for the Properties of Water and Steam. *Release on the IAPWS Formulation 2011 for the Thermal Conductivity of Ordinary Water Substance*. IAPWS R15-11, Plzeň. 2011.

[17] International Association for the Properties of Water and Steam. *Revised Release on the IAPWS Formulation 1995 for the Thermodynamic Properties of Ordinary Water Substance for General and Scientific Use*. IAPWS R6-95(2018), Prague. 2018.

[18] Arthur Jaffe, Edward Witten. *Quantum Yang-Mills Theory*. Clay Mathematics Institute, Millennium Prize Problem statement, `https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf`.

[19] O. A. Ladyzhenskaya. *New equations for the description of the motions of viscous incompressible fluids, and global solvability for their boundary value problems*. Trudy Mat. Inst. Steklov. 102, 85–104. 1967.

[20] O. A. Ladyzhenskaya. *On modifications of the Navier–Stokes equations for large gradients of the velocities*. Zap. Nauchn. Sem. LOMI 7, 126–154. 1968.

[21] O. A. Ladyzhenskaya. *The Mathematical Theory of Viscous Incompressible Flow*. Second English edition, translated by R. A. Silverman and J. Chu, Gordon and Breach. 1969.

[22] J. Leray. *Sur le mouvement d'un liquide visqueux emplissant l'espace*. Acta Mathematica 63, 193–248, doi:10.1007/BF02547354; translation by R. E. Terrell, arXiv:1604.02484. 1934.

[23] C. Lienstromberg, S. Schiffer, R. Schubert. *A variational approach to the Navier–Stokes equations with shear-dependent viscosity*. arXiv:2312.03546v2. 2025.

[24] C.-L. Navier. *Mémoire sur les lois du mouvement des fluides*. Mémoires de l'Académie royale des sciences de l'Institut de France 6, 389–440, read 18 March 1822. 1827.

[25] A. M. Odlyzko. *The 10^{22}-nd zero of the Riemann zeta function*. In Dynamical, Spectral, and Arithmetic Zeta Functions, Contemporary Mathematics 290, American Mathematical Society, 139-144. 2001.

[26] OpenAI. *Finite time blowup for the Euler equation*. `https://cdn.openai.com/pdf/315b36cd-ec98-4023-8342-93345194ece1/euler.pdf`. 2026.

[27] OpenAI. *NavierStokesAndEuler*, Lean 4 formalizations of the Navier-Stokes and Euler blowup results. `https://github.com/openai/NavierStokesAndEuler`. 2026.

[28] OpenAI. *Finite time blowup for Navier-Stokes*. `https://cdn.openai.com/pdf/32d9f210-8b73-45e0-91bc-82a30aef8a9a/navier-stokes.pdf`. 2026.

[29] D. Platt, T. Trudgian. *The Riemann hypothesis is true up to* $3 \cdot 10^{12}$. Bulletin of the London Mathematical Society 53(3), 792-797. 2021.

[30] D. Quigg. *Crystallography*. orior research papers, https://github.com/dstroy0/orior. 2026.

[31] D. Quigg. *The Instruments*. orior research papers, https://github.com/dstroy0/orior. 2026.

[32] Bernhard Riemann. *Handwritten working manuscript, Nachlass Cod. Ms. Riemann 3, Rechenseiten*. Niedersaechsische Staats- und Universitaetsbibliothek Goettingen, Cod. Ms. Riemann 3; scan page 00000038.jpg. 1859.

[33] NYU Shanghai RITS. *OpenAI Claims a Navier-Stokes Proof, Amid a Dispute Over Credit*. https://rits.shanghai.nyu.edu/ai/openai-navier-stokes-proof-credit-dispute/. 2026.

[34] G. G. Stokes. *On the theories of the internal friction of fluids in motion, and of the equilibrium and motion of elastic solids*. Transactions of the Cambridge Philosophical Society 8, 287–319; reprinted in Mathematical and Physical Papers, vol. I, Cambridge, 1880, 75–129. 1845.

[35] Andrew Wiles. *The Birch and Swinnerton-Dyer Conjecture*. Clay Mathematics Institute, Millennium Prize Problem statement, https://www.claymath.org/wp-content/uploads/2022/05/birchswin.pdf.

[36] Bernhard Riemann, translated by David R. Wilkins. *On the Number of Prime Numbers less than a Given Quantity*. Preliminary version, December 1998, of Ueber die Anzahl der Primzahlen unter einer gegebenen Grosse, Monatsberichte der Berliner Akademie, November 1859.